

AGNIBH DASGUPTA

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Education

2021 - 2026 PhD, Computing & Information Science, **University of Nebraska, Omaha**
2018 - 2020 Master of Science, Computer Science, **Utah State University**
2013 - 2017 Bachelor of Technology, Computer Science, **West Bengal University of Technology**

Publications

- 2025 **Watermarking Language Models through Language Models (IEEE TAI)** [🔗](#)
• Built a prompt-based **LLM watermarking** framework without modifying model weights
• Achieved **88%+ watermark detection accuracy** across 25 **open-source LLMs**
- 2024 **Book Chapter, Imaging Science** [🔗](#)
• Improved the scalability of our attention-based watermarking under heavy compound noise
- 2024 **Leveraging Artificial Intelligence (AI) to Enhance CS Instruction (FIE)** [🔗](#)
• Utilized multi-modality in 3D CNN models for mood detection in K-12 classroom videos
- 2023 **Robust Image Watermarking based on Cross-Attention and ID Learning (CSCI)** [🔗](#)
• Designed an attention-based robust image watermarking scheme using **vision transformers**
• Trained a robust invariant domain using **self-supervised learning**
- 2022 **Perspective transformation layer (CSCI)** [🔗](#)
• Proposed a lightweight layer that learns 2D projections of 3D views for multiple viewpoints

Present and Past Research

- PRESENT **Invariant Representation Learning in LLMs for Model Attribution** (*under review*)
• Analyzed features from **10 open-source LLMs** using **PCA** and custom similarity measures
• Found model-specific representation patterns that enable **reliable model attribution**
- PRESENT **Invariant Feature Learning with Auto-Augmentation for Watermarking** (*under review*)
• Combined **CLIP image-text** features with a **learned augmentation module** evaluated on **ImageNet** and **Visual Genome** datasets
• Exceeded state-of-the-art watermark robustness to camera and environmental distortions
- 2025 **Utilization of SAM2 Model for the Segmentation of Retinal Vascular Leakage on Ultra-wide-field Fluorescein Angiography in Non-infectious Retinal Vasculitis (AAO)**
• Trained a classifier on patches of large optic scans to accurately detect areas of leakage
- 2022 **Sentiment Analysis on Twitter data on Electric Vehicles**
• Analyzed sentiment with **lexical** and **Word2Vec-LSTM** approaches on scraped twitter data
- 2020 **Deep Q Learning applied to stock trading** [🔗](#)
• Built a **reinforcement learning trading system** using auto-scraped S&P index data

Reviewer Service

2022 - PRESENT ACCV, Scientific Reports - Nature, Springer Nature

Teaching Experience

2022 - 2023 **Intern Supervisor**, University of Nebraska at Omaha
2023 - 2025 **Graduate Teaching Assistant**, University of Nebraska at Omaha
Courses: Intro to Python, Capstone
2018 - 2020 **Graduate Teaching Assistant**, Utah State University
Courses: Programming Languages, Intro to C++, Intro to AI

Technical Skills

PROGRAMMING & SYSTEMS:	Python, Bash, Linux, MATLAB, Git, JAX, CUDA
DEEP LEARNING FRAMEWORKS:	PyTorch, TensorFlow, Transformers, LLMs, NLP
DISTRIBUTED & OPTIMIZATION:	DeepSpeed, PyTorch DDP, tf.distribute, Slurm/HPC

Motivation Statement - Humboldt Postdoctoral Fellowship

Hello,

I am a PhD student graduating in May 2026, and my research centers on invariant representation learning and watermarking in vision and language domains, with an emphasis on robustness and generalization. I study how latent representations behave under perturbation and how invariant representations can be explicitly learned to support robust decision-making.

Some of my relevant published work includes, developing a cross-attention-based watermarking method and an adversarially trained invariant representation space that remains stable under severe compound augmentation. I have also worked extensively with several open-source LLMs, designing a prompt-driven watermarking framework that involved fine-tuning and distillation across multiple state-of-the-art LLM families. More recently, my under-review research includes 1) a zero-watermarking framework, which utilizes learned adversarial augmentations and multimodal representation using CLIP to train an invariant representation space, and 2) a study of representation invariance in LLMs using custom layer-wise metrics to cluster latent representations across LLM families for model attribution and behavior analysis.

I also have some applied AI experience in, 1) using deep reinforcement learning for stock market trading as part of my master's thesis, 2) multimodal sentiment analysis of K-12 classroom videos, in collaboration with education researchers, with the goal of providing feedback to instructors for enhanced education, 3) detecting areas of leakage in ocular scans in collaboration with, and in assistance to medical practitioners.

I have seven years of teaching experience as a TA, and I have mentored high school interns in my lab by interviewing and shortlisting candidates, supervising their research, and helping them contribute to ongoing projects.

Going forward, I'm interested in studying how invariant and generalizable structures emerge across vision and language models, and how these shared representations can be exploited to improve robustness and explainability in multimodal AI systems. I am motivated by the need for AI models, particularly going into the future, to not only perform well, but do so reliably, transparently and ethically across diverse conditions and applications, and I aim to contribute principled methods that move the field toward more trustworthy, interpretable and ethical AI.

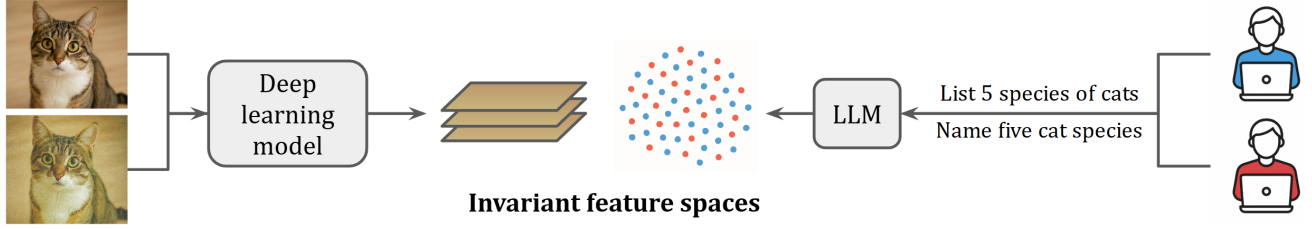
With respect to the CAI group, I am interested in working on projects like the ones titled 'Identifying Explanation-Worthy Features in Large Language Models' and 'Gaze-Driven Foveation for Efficient VLMs' in the open projects section.

I would welcome the opportunity to discuss my research and potential collaboration. Thank you for your consideration.

Sincerely,

Agnibh

Robust Invariant Representation Learning Across Modalities



Modern AI systems succeed not merely because they fit large datasets, but because they learn *invariant representations*. These are stable features that preserve meaning and remain consistent despite significant variation in inputs. Such representations enable a vision model to recognize an object under different lighting or pose, and allow a language model to understand paraphrases expressed with entirely different surface forms. Despite their central role in generalization, these invariant structures remain among the least understood aspects of deep learning, especially as models expand across modalities that include images, text, audio, and video.

My research investigates how invariant representations emerge in modern neural networks, how they differ across modalities, and how they can be *explicitly* learned and *operationalized* for robustness-critical applications such as watermarking. The motivation is grounded in two empirical observations. First, large-scale self-supervised vision models such as DINO and iBOT exhibit strong invariance to geometric and photometric transformations through alignment of augmented views [1], [2]. Second, transformer language models show layerwise semantic convergence, with deeper layers encoding paraphrase-stable representations that collapse meaning-preserving variants into compact latent neighborhoods [3], [4]. Yet in both domains, such invariances predominantly arise implicitly as artifacts of current objectives, with only limited forms of explicit invariance appearing in specialized architectures or training regimes.

This implicitness creates vulnerabilities. Vision models robust to standard augmentations can fail under unexpected but semantics-preserving perturbations, a limitation documented in robustness benchmarks such as AugLy and DomainBed [5], [6]. Language models exhibit similar fragility, where paraphrasing and stylistic shifts degrade performance on tasks that assume semantic consistency. As multimodal systems become foundational for modern AI, these failures compound. Robustness can no longer be treated as purely domain specific. A unified hypothesis space for invariance is required to reason about perturbations across modalities, recognizing that different modalities may exhibit partially aligned (but not fully shared) invariant structures.

My work addresses this gap through three integrated directions. *First*, I map the invariant subspaces that large vision and language models already learn, revealing where semantic stability naturally arises and where it breaks. *Second*, I develop mechanisms for training models toward these stable features, including learned perturbation curricula and multimodal alignment strategies. *Third*, I use watermarking as a functional stress test for invariance. A watermark that survives paraphrasing, camera recapture, or cross-modal transformation reflects true representational stability, while failures highlight gaps in the invariant geometry.

Together, these directions establish a cross-modal research agenda that connects invariance, robustness, and provenance. By grounding robustness-critical tasks such as watermarking in the geometry of invariant latent spaces, my research aims to contribute empirical and methodological evidence toward a principled account of how invariant representations can be leveraged in building more reliable AI systems that are robust and empirically verifiable at the representation level across modalities.

I. PRIOR RESEARCH

Perspective Transformation Layer. An early line of my work introduced the Perspective Transformation Layer (PTL) [7], a differentiable module that learns task-relevant geometric transformations within the network’s forward pass. Rather than relying on fixed, hand-crafted augmentation pipelines, PTL internalizes perspective warping as a learnable operator, allowing the model to adaptively choose which viewpoint changes improve downstream accuracy. PTL established a core principle that informs my later work: robustness improves when transformation handling is built *into the model’s representational process*, not left entirely to external data augmentation.

Robust Image Watermarking based on Cross-Attention and Invariant Domain Learning. Building on this insight, I developed a representation-level watermarking framework that explicitly ties watermark extraction to invariant visual features [8]. The method co-trains an encoder to map augmented views of an image to a stable latent space while a cross-attention mechanism localizes semantically meaningful regions for watermark binding. By grounding the watermark in the model’s invariant features rather than pixel-level artifacts, the system achieves strong robustness to noise, compression, cropping, and other semantic-preserving distortions.

Watermarking Language Models Through Language Models. I then extended this representational perspective to large language models through a black-box watermarking framework [9]. Instead of modifying logits or internal weights, this approach uses one model to prompt another to generate watermarked text and a separate model to detect it. Because watermarking operates entirely at the behavioral level, it generalizes across architectures and remains effective under paraphrasing, fine-tuning, and lightweight distillation. Evaluations across more than twenty Prompting–Marking–Detecting model combinations (including GPT-4o, Mistral, DeepSeek, and LLaMA families) show that watermark signals can persist even when the underlying model changes.

II. ONGOING RESEARCH

TIACam: Learned Perturbations and Text-Anchored Invariance for Camera-Robust Watermarking. My current work in vision develops TIACam, a framework that treats watermarking as an invariance-learning problem rather than an embedding problem. This work extends my earlier invariant-domain watermarking framework by replacing fixed augmentation recipes with a learned perturbation mechanism that discovers camera-like distortions through differentiable operators. By letting the model adversarially generate the transformations it struggles with, TIACam grounds invariance in the model’s own failure modes rather than in human-designed heuristics. A text-anchored invariant learner further links visual representations to stable semantic descriptions using a CLIP backbone, enabling robustness that spans both geometric and photometric transformations as well as cross-modal cues. TIACam therefore represents the vision realization of my broader research agenda: explicitly shaping invariant feature spaces and using watermarking as the functional stress test that validates them.

Invariant Layer Discovery and Latent Behavior Signatures in LLMs. In parallel, I extend this invariance perspective to language by investigating whether large language models contain latent structures analogous to invariant visual features. Building on prior work in black-box LLM watermarking, this project shifts the focus from surface-level patterns to the internal geometry of model activations. By analyzing how layer representations behave under paraphrasing, summarization, and stylistic reformulation, the work identifies subspaces that remain stable across meaning-preserving transformations and differentiate model families through their behavioral regularities. These invariant subspaces provide a foundation for representation-level watermarking in language, where watermark signals can be tied to stable latent features rather than token choices. More broadly, the project serves as the language instantiation of my cross-modal agenda, revealing how invariance emerges in LLMs and how it can be operationalized for robustness-critical tasks.

III. OTHER RESEARCH

In a preliminary K–12 education paper [10], I helped design a 3D convolutional model to analyze classroom video for coarse mood and engagement patterns, with the long-term goal of providing feedback to teachers about classroom dynamics. This work served as a proof of concept and a foundation for future, more detailed studies on human-centered AI support for instruction.

In a medical imaging collaboration [11], I developed preliminary models to localize retinal vascular leakage by drawing bounding boxes over candidate regions, and experimented with generating one type of retinal scan from another under limited data conditions. This project explored how foundation-model-style representations and generative mappings could be adapted to highly specialized clinical imagery.

IV. FUTURE RESEARCH DIRECTION

Central Research Question:

What invariant structures arise and are shared across modalities, and how can these structures be explicitly learned and exploited to make AI systems inherently robust, interpretable, and verifiable?

This question arises at a pivotal moment for modern AI. As models increasingly operate over images, text, audio, video, and other data streams, robustness can no longer be treated as a property of a single modality. Multimodal systems depend on stable, meaning-preserving representations that remain consistent across heterogeneous inputs, yet current architectures rely on invariances that emerge only implicitly and often fail under simple, semantics-preserving perturbations. At the same time, real-world concerns such as provenance, attribution, and safe generative modeling demand methods whose behavior can be probed and evaluated at the representation level, rather than only via surface outputs.

My future work will aim to address these challenges by studying and shaping the invariant geometry that underlies multimodal learning. I aim to (1) characterize the invariant subspaces that different modalities naturally share, revealing how models encode stable structure across visual and linguistic domains, (2) develop explicit training mechanisms, including learned perturbation curricula and cross-modal alignment objectives, that reliably produce stable and coherent invariant features, and (3) operationalize these invariant spaces for robustness-critical applications such as watermarking, where survival under paraphrasing, editing, and recapture provides a direct and measurable test of representational stability.

Together, these directions pursue a unified goal: to build AI systems whose reliability comes from principled invariant representations rather than from task-specific heuristics. By grounding robustness, interpretability, and provenance in the structure of multimodal latent spaces, my research aims to contribute methods that address foundational scientific questions while responding directly to emerging needs in trustworthy and safe generative AI.

REFERENCES

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- [2] J. Zhou, C. Wei, *et al.*, “ibot: Image bert pre-training with online tokenizer,” *arXiv preprint arXiv:2111.07832*, 2021.
- [3] K. Ethayarajh, “How contextual are contextualized word representations?,” in *Proceedings of EMNLP*, 2019.
- [4] I. Tenney, D. Das, and E. Pavlick, “Bert rediscovers the classical nlp pipeline,” in *Proceedings of ACL*, 2019.
- [5] X. Li *et al.*, “Augly: Data augmentations for robustness,” in *NeurIPS Datasets and Benchmarks*, 2021.
- [6] I. Gulrajani and D. Lopez-Paz, “In search of lost domain generalization,” in *International Conference on Learning Representations (ICLR)*, 2021.
- [7] N. Khatri, A. Dasgupta, Y. Shen, X. Zhong, and F. Y. Shih, “Perspective transformation layer,” in *2022 International Conference on Computational Science and Computational Intelligence (CSCI)*, pp. 1395–1401, 2022.
- [8] A. Dasgupta and X. Zhong, “Robust image watermarking based on cross-attention and invariant domain learning,” in *2023 International Conference on Computational Science and Computational Intelligence (CSCI)*, pp. 1125–1132, 2023.
- [9] A. Dasgupta, A. A. Tanvir, and X. Zhong, “Watermarking language models through language models,” *IEEE Transactions on Artificial Intelligence*, pp. 1–10, 2025.
- [10] H. Yoon, X. Zhong, A. Dasgupta, G. Nugent, and G. Trainin, “Leveraging artificial intelligence (ai) to enhance computer science instruction,” in *2024 IEEE Frontiers in Education Conference (FIE)*, pp. 1–5, 2024.
- [11] X. Xing, I. Karaca, A. Akhavanrezaayat, S. Badrloo, Q. D. Nguyen, and M. Subramaniam, “Mae-sam2: Mask autoencoder-enhanced sam2 for clinical retinal vascular leakage segmentation,” 2025.

List of Publications: Agnibh Dasgupta

Peer-Reviewed Journal & Conference Papers

- **Dasgupta, A.**, Tanvir, A. A., & Zhong, X. (2025). **Watermarking Language Models Through Language Models**. *2024 International Conference on Computational Science and Computational Intelligence (CSCI)*. Published by IEEE Xplore. doi: 10.1109/CSCI63914.2024.00115
<https://ieeexplore.ieee.org/document/11146861>
- **Dasgupta, A.**, & Zhong, X. (2025). **Enhanced image watermarking through cross-attention and noise-invariant domain learning**. *Imaging Science: Computer Vision, Image and Signal Processing, Pattern Recognition*, 1–15. Published by De Gruyter.
<https://ieeexplore.ieee.org/abstract/document/11052895>
- Yoon, H. J., Zhong, X., **Dasgupta, A.**, Nugent, G., & Trainin, G. (2024). **Leveraging Artificial Intelligence (AI) to Enhance Computer Science Instruction**. *2024 IEEE Frontiers in Education Conference (FIE)*, 1–5.
<https://ieeexplore.ieee.org/document/10893354>
- **Dasgupta, A.**, & Zhong, X. (2023). **Robust image watermarking based on cross-attention and invariant domain learning**. *2023 International Conference on Computational Science and Computational Intelligence (CSCI)*, 614–619.
<https://ieeexplore.ieee.org/document/10590390>
- Khatri, N., **Dasgupta, A.**, Shen, Y., Zhong, X., & Shih, F. Y. (2022). **Perspective transformation layer**. *2022 International Conference on Computational Science and Computational Intelligence (CSCI)*, 1402–1407.
<https://ieeexplore.ieee.org/document/10216469>

Theses & Technical Reports

- **Dasgupta, A.** (2022). *Learning an Invariant Domain for Image Watermarking* [Doctoral Research/Technical Report]. University of Nebraska at Omaha.
- **Dasgupta, A.** (2020). *Deep Q learning applied to stock trading* [Master's Thesis]. Utah State University.



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TRANSCRIPT AS ON 21-09-2017

Name	AGNIBH DASGUPTA					Roll No	10200113005				
Course	B.TECH (COMPUTER SCIENCE & ENGINEERING)					Reg No	131020110005 OF 2013-2014				
Subject Code	Subjects Offered	Letter Grade	Points	Credit	Credit Points	Subject Code	Subjects Offered	Letter Grade	Points	Credit	Credit Points
HU-101	ENGLISH LANGUAGE & TECHNICAL COMMUNICATION	C	6	2.0	12.0	CS201	BASIC COMPUTATION & PRINCIPLES OF COMPUTER PROGRAMMING	E	9	4.0	36.0
PH-101	PHYSICS-I	C	6	4.0	24.0	CH201	CHEMISTRY-I	B	7	4.0	28.0
M-101	MATHEMATICS-I	C	6	4.0	24.0	M201	MATHEMATICS-II	C	6	4.0	24.0
ME-101	ENGINEERING MECHANICS	A	8	4.0	32.0	ME201	ENGINEERING THERMODYNAMICS & FLUID MECHANICS	D	5	4.0	20.0
ES-101	BASIC ELECTRICAL & ELECTRONIC ENGINEERING-I	D	5	4.0	20.0	ES201	BASIC ELECTRICAL & ELECTRONIC ENGINEERING-II	C	6	4.0	24.0
PH191	PHYSICS-I	A	8	2.0	16.0	CS291	BASIC COMPUTATION & PRINCIPLES OF COMPUTER PROGRAMMING LAB	E	9	2.0	18.0
ES191	BASIC ELECTRICAL & ELECTRONIC ENGINEERING-I	A	8	2.0	16.0	CH291	CHEMISTRY-I	C	6	2.0	12.0
ME192	WORKSHOP PRACTICE	A	8	3.0	24.0	ES291	BASIC ELECTRICAL & ELECTRONIC ENGINEERING-II LAB	A	8	2.0	16.0
HU181	LANGUAGE LABORATORY	A	8	1.0	8.0	ME291	BASIC ENGG DRAWING & COMPUTER GRAPHICS	E	9	3.0	27.0
XC181	EXTRA CURRICULAR ACTIVITIES (NSS/NCC/NSO ETC.)	A	8	1.0	8.0						
	Total			27.0	184.0		Total			29.0	205.0
PASSED 2014 SGPA1: 6.81						PASSED 2014 SGPA2: 7.07 YGPA1: 6.95					
HU-301	VALUES & ETHICS IN PROFESSION	B	7	3.0	21.0	M(CS)401	NUMERICAL METHODS	C	6	2.0	12.0
PH-301	PHYSICS-II	D	5	4.0	20.0	M401	MATHEMATICS-3	D	5	4.0	20.0
CH-301	BASIC ENVIRONMENTAL ENGINEERING & ELEMENTARY BIOLOGY	C	6	3.0	18.0	CS401	COMMUNICATION ENGG & CODING THEORY	C	6	3.0	18.0
CS-301	ANALOG & DIGITAL ELECTRONICS	D	5	3.0	15.0	CS402	FORMAL LANGUAGE & AUTOMATA THEORY	D	5	4.0	20.0
CS-302	DATA STRUCTURE & ALGORITHM	B	7	4.0	28.0	CS403	COMPUTER ARCHITECTURE	C	6	4.0	24.0
CS-303	COMPUTER ORGANIZATION	D	5	4.0	20.0	HU481	TECHNICAL REPORT WRITING & LANGUAGE LAB PRACTICE	A	8	2.0	16.0
PH391	PHYSICS-2	B	7	2.0	14.0	M(CS)491	NUMERICAL METHODS	A	8	1.0	8.0
CS391	ANALOG & DIGITAL ELECTRONICS	A	8	2.0	16.0	CS491	COMMUNICATION ENGG & CODING THEORY	E	9	2.0	18.0
CS392	DATA STRUCTURE & ALGORITHM	O	10	2.0	20.0	CS492	SOFTWARE TOOLS	O	10	2.0	20.0
CS393	COMPUTER ORGANISATION	O	10	2.0	20.0	CS493	COMPUTER ARCHITECTURE	O	10	2.0	20.0
	Total			29.0	192.0		Total			26.0	176.0
PASSED 2015 SGPA3: 6.62						PASSED 2015 SGPA4: 6.77 YGPA2: 6.69					
HU501	ECONOMICS FOR ENGINEERS	D	5	3.0	15.0	HU601	PRINCIPLES OF MANAGEMENT	C	6	2.0	12.0
CS501	DESIGN & ANALYSIS OF ALGORITHM	C	6	4.0	24.0	CS601	DATA BASE MANAGEMENT SYSTEM	C	6	3.0	18.0
CS502	MICROPROCESSORS & MICROCONTROLLERS	D	5	4.0	20.0	CS602	COMPUTER NETWORKS	C	6	3.0	18.0
CS503	DISCRETE MATHEMATICS	C	6	3.0	18.0	CS603	OPERATING SYSTEM	D	5	3.0	15.0
CS504D	OBJECT ORIENTED PROGRAMMING	B	7	4.0	28.0	CS604A	INFORMATION THEORY & CODING	B	7	3.0	21.0
CS591	DESIGN & ANALYSIS OF ALGORITHM	E	9	2.0	18.0	CS605C	MULTIMEDIA TECHNOLOGY	C	6	4.0	24.0
CS592	MICROPROCESSORS & MICROCONTROLLERS	O	10	2.0	20.0	CS691	DATA BASE MANAGEMENT SYSTEM LAB	B	7	2.0	14.0
CS593	PROGRAMMING PRACTICES USING C++	A	8	2.0	16.0	CS692	NETWORK LAB	B	7	2.0	14.0
CS594D	OBJECT ORIENTED PROGRAMMING	A	8	2.0	16.0	CS693	OPERATING SYSTEM LAB	E	9	2.0	18.0
	Total			26.0	175.0	CS681	SEMINAR	O	10	2.0	20.0
PASSED 2017 SGPA5: 6.73						Total 26.0 174.0					
PASSED 2017 SGPA6: 6.69 YGPA3: 6.71						PASSED 2017 SGPA6: 6.69 YGPA3: 6.71					
CS701	SOFTWARE ENGG	B	7	3.0	21.0	HU801A	ORGANISATIONAL BEHAVIOUR	C	6	2.0	12.0
CS702	COMPILER DESIGN	C	6	3.0	18.0	CS801D	CRYPTOGRAPHY & NETWORK SECURITY	B	7	3.0	21.0
CS703C	ARTIFICIAL INTELLIGENCE	C	6	3.0	18.0	CS802E	E-COMMERCE(IT)	A	8	3.0	24.0
CS704E	MOBILE COMPUTING	C	6	3.0	18.0	CS891	DESIGN LAB / INDUSTRIAL PROBLEM RELATED PRACTICAL TRAINING	O	10	4.0	40.0
CS705A	INTERNET TECHNOLOGY	C	6	3.0	18.0	CS892	PROJECT-2	O	10	6.0	60.0
HU781	GROUP DISCUSSION	A	8	2.0	16.0	CS893	GRAND VIVA	E	9	3.0	27.0
CS791	SOFTWARE ENGG. LAB	O	10	2.0	20.0		Total			21.0	184.0
CS793C	ARTIFICIAL INTELLIGENCE	E	9	2.0	18.0	PASSED 2017 SGPA8: 8.76 YGPA4: 8.06 DGPA: 7.16					
CS795A	INTERNET TECHNOLOGY	O	10	2.0	20.0	PASSED 2017 SGPA8: 8.76 YGPA4: 8.06 DGPA: 7.16					
CS792	INDUSTRIAL TRAINING	O	10	2.0	20.0						
CS794	PROJECT-1	A	8	2.0	16.0						
	Total			27.0	203.0						
PASSED 2017 SGPA7: 7.52											
COLLEGE/INSTITUTION : KALYANI GOVERNMENT ENGINEERING COLLEGE											

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Controller of Examinations (OSD)
Maulana Abul Kalam Azad University
of Technology, West Bengal

Registrar / Controller of Examinations
(Formerly known as West Bengal University of Technology)

Record of: Agnibh Dasgupta

Student ID: A02292865

SSN: *****1540

Date of Birth: 12-DEC

Course Level: Graduate

Current Program
Master of Science
Major : Computer Science

SUBJ	NO.	COURSE TITLE	CRED	GRD	PTS	R
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INSTITUTION CREDIT:

Fall 2018

CS	5050	ADV ALGORITHMS	3.00	B+	9.99	
CS	5800	INTRO TO DATABASE SYSTEMS	3.00	A	12.00	
CS	6680	COMPUTER VISION	3.00	A-	11.01	
CS	6900	SEMINAR	1.00	P	0.00	
USU	6900	RESEARCH INTEGRITY	0.00	P	0.00	
USU	7920	TEACHING ASSISTANT WORKSHOP	0.00	P	0.00	
Ehrs: 10.00 GPA-Hrs: 9.00 QPts: 33.00 GPA: 3.66						

Good Standing

Spring 2019

CS	5665	INTRODUCTION TO DATA SCIENCE	3.00	B	9.00	
CS	6890	ST: REINFORCEMENT LEARNING	3.00	A-	11.01	I
STAT	6910	ST:DEEP LEARNING THEORY&APP	3.00	B+	9.99	
Ehrs: 9.00 GPA-Hrs: 9.00 QPts: 30.00 GPA: 3.33						

Good Standing

Fall 2019

CS	6890	ST: MACHINE INTEL CLEAN ENERGY	3.00	A	12.00	I
CS	6950	DIRECTED READINGS	3.00	IF	0.00	
Ehrs: 3.00 GPA-Hrs: 3.00 QPts: 12.00 GPA: 4.00						

Good Standing

Spring 2020

IN PROGRESS WORK						
CS	6970	THESIS AND RESEARCH	6.00	IN PROGRESS		

In Progress Credits 6.00

***** CONTINUED ON NEXT COLUMN *****

***** TRANSCRIPT TOTALS *****	
	Earned Hrs GPA Hrs Points GPA
TOTAL INSTITUTION	22.00 21.00 75.00 3.57
TOTAL TRANSFER	0.00 0.00 0.00 0.00
OVERALL	22.00 21.00 75.00 3.57
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Grading Scale

Grade	Points	Grade	Points	Notes
A	4.00	P	0.00	Passing
A-	3.67	NF	0.00	Failed Due to Non-Participation
B+	3.33	AU	0.00	Audit (performance not assessed)
B	3.00	NGR	0.00	No Grade Reported
B-	2.67	I	0.00	Incomplete
C+	2.33	W	0.00	Withdrawal
C	2.00	SP	0.00	Satisfactory Progress
C-	1.67	SC	0.00	Satisfactory Completion
D+	1.33	CR	0.00	Credit(s)
D	1.00			
F	0.00			

'I' plus a companion grade has no GPA-Hrs or EHRS. It indicates the student has up to one calendar year to finish the work. 'W' plus a companion grade carries no grade point value. It indicates the student withdrew after 60% of the course; this is not an unofficial withdrawal.

Symbols

EHRS	Earned Hours
AHRS	Attempted Hours
QPTS	Quality Points
GPA-Hrs	Hours Used in GPA Calculations
ST	Special Topics Course
S	Service Learning Course
H	Honors Course
@	Remedial Course - No GPA-Hrs or EHRS
R	Academic Renewal - No GPA-Hrs or EHRS
V	Honor Code Violation - Academic Dishonesty (may be removed by petition after one year)

Repeated Courses appear in the far-right column marked titled "R" with the following notations:

- E Excluded from GPA-Hrs and EHRS
- I Included in GPA-Hrs and EHRS
- A Included in GPA-Hrs and excluded from EHRS

Semester Conversion and Graduation Requirements

- ♦ USU converted to semesters beginning Fall 1998. Transfer credit has been converted to semester credits.
- ♦ The baccalaureate degree requires a minimum of 120 semester credits and a minimum grade point average of 2.00.
- ♦ All courses displayed as semester credit

Course Numbering

Prior to June 1971

001-099: Lower Division; 100-199: Upper Division (may be graduate if bachelor's degree has been awarded); 200 and above: Graduate.

June 1971 - August 1998

001-099	Remedial Courses - No GPA-Hrs or EHRS
100-299	Lower Division
300-499	Upper Division
500-599	Advanced Upper Division (or graduate credit with Dean approval)
600-799	Graduate Courses

September 1998 - present

0010-0999	Remedial Courses - No GPA-Hrs or EHRS
1000-2999	Lower Division
3000-4999	Upper Division
5000-5999	Advanced Upper Division (or graduate credit with Dean approval)
6000-7999	Graduate Courses

University Studies Codes (Beginning Fall 1998)

CL1	Communications Literacy 1	CL2	Communications Literacy 2
CI	Communications Intensive	QL	Quantitative Literacy
QI	Quantitative Intensive		

Breadth Courses

BAI	American Institutions
BCA	Creative Arts
BHU	Humanities
BLS	Life Sciences
BPS	Physical Sciences
BSC	Sciences
BSS	Social Sciences

Depth Courses

DHA	Humanities & Creative Arts
DSS	Social Sciences
DSC	Life & Physical Sciences

Confidentiality of Records

In compliance with the Family Educational Rights and Privacy Act, this transcript must not be released to a third party without written authorization of the student.

Utah State University is governed by the Utah State Board of Regents and is fully accredited by the Northwest Commission on Colleges and Universities. See the university catalog for details about professional and specialized accreditations.